VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Pilot Vehicle Interface (PVI)

AFLCMC/WA

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1.0 Overview

The Pilot Vehicle Interface (PVI) connects the pilot with the platform and the Agile Mission Suite (AMS) Government Reference Architecture (GRA) Mission System.

1.1 Purpose

The PVI Volume describes the process a Mission System Integrator (MSI) should follow when developing a PVI to integrate with an AMS GRA Mission System.

1.2 Scope

The PVI Volume aids the MSI in designing the specific PVI implementation for a platform.

1.3 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or Minimum Procurable Unit (MPU) provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.3.1** for a list of informative references for the PVI Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	Software Architecture Volume	14.0

Table 1.3.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the PVI and Related Entities.

2.0 Pilot Vehicle Interface (PVI)

The tactical PVI is the mechanism the pilot uses to interact with the aircraft and the AMS GRA Mission System. Though PVI typically includes a wide variety of hardware and software components, this section primarily focuses on PVI as it relates to displaying graphical formats on glass.

The tactical PVI is external to the AMS GRA boundary. Therefore, AMS GRA does not dictate the architecture of the PVI software. However, to enable design and analysis of various system architectures, it is helpful to have a common definition of PVI. **Figure 2.0.1** below shows a notional PVI architecture interacting with AMS GRA software. It is important to note that the data model adapter isolates the rest of the PVI from the messaging protocol. No matter which protocol is used for messaging between the PVI and Mission System Skills or Services, the data model adapter needs to be able to convert between internal data types and messaging data types. It is the only portion of PVI that is required to change when an applicable message definition changes.

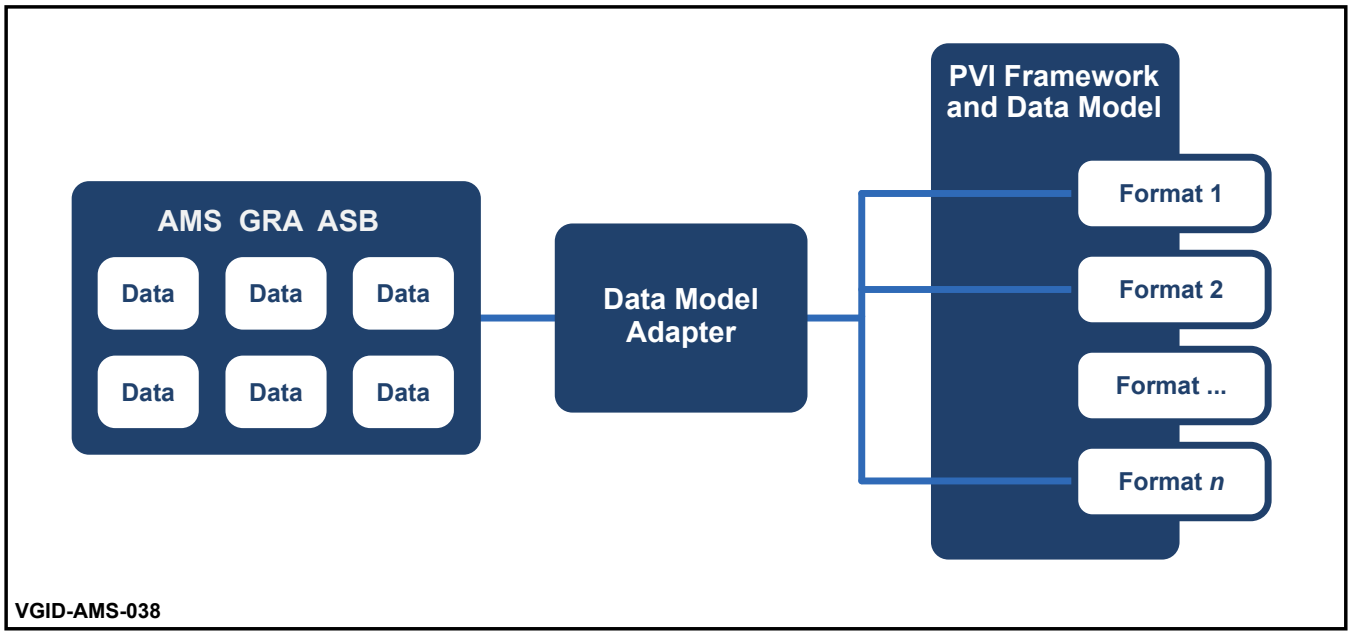


Figure 2.0.1: This Diagram Shows a Notional PVI Software Architecture that Interfaces with an AMS GRA Mission System.

2.1 Interface Decision Tree

The choice of AMS GRA-to-PVI architecture is platform dependent. This choice poses two main questions:

1. Is it feasible for an adopting program to route data through a PVI Skill in order to provide the isolation and simplicity of an isolated PVI?
2. Is it feasible for an adopting program to use the Abstract Service Bus (ASB) in order to maintain a fully open architecture and promote the sharing of data?

When determining feasibility of a solution, each adopting program needs to consider:

- The required level of alignment with key tenets of AMS GRA
- The amount of time and budget available for implementation and testing
- Software interface or physical hardware restrictions related to protocols, routing, and processing

For example, if existing AMS GRA software already outputs data onto the Avionics Backplane Bus (ABB) or ASB with the ability to translate the data into a format the existing PVI software already accepts, then the Isolated PVI Pattern, shown in **Figure 2.1.1**, is likely the best choice. However, if the bus would require the addition of a new message, and time is of the utmost importance, the program may be willing to accept a lower level of alignment with the key tenets of AMS GRA in order to meet strict timelines.

The decision tree in **Figure 2.1.1** shows which approach is favored based on the underlying software implementation and the priorities of the adopting program. The key variables considered during construction of **Figure 2.1.1** were:

- The impact to the overall system
- The number of components impacted
- The time to implement within a PVI architecture
- The time to implement within AMS GRA
- The openness of the data being integrated
- Bandwidth and latency
- Severability
- Testability

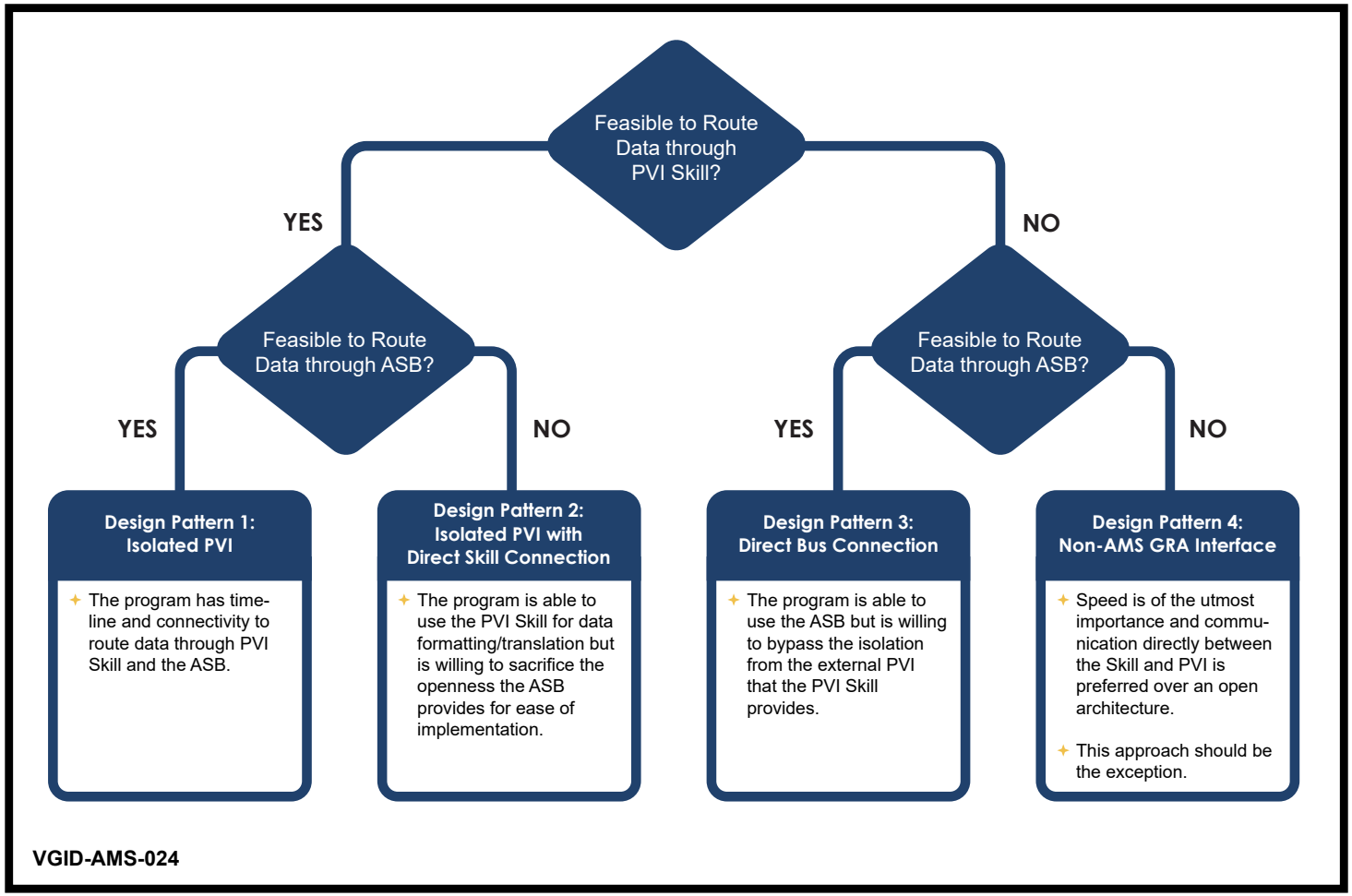


Figure 2.1.1: This Interface Approach Decision Tree Shows How to Decide the Preferred PVI Design Pattern to Select for an Adopting Program.

2.2 Advanced PVI Skills

Many AMS GRA Skills provide partial or full automation within the Mission System, which reduces manual pilot interaction with the PVI. Some Skills or decisions cannot or should not be automatic based on Rules of Engagement (ROE) or safety of flight. AMS GRA takes a significant leap forward with respect to PVI by defining Skills to reduce pilot workload and aid decision making. These new PVI Skills aim to provide the pilot with actionable knowledge rather than only providing traditional data.

3.0 Compliance

No compliance requirements have been identified for an MSI PVI implementation. It is the responsibility of the MSI to develop a PVI solution that is compatible with the rest of AMS GRA.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
ASB	Abstract Service Bus
GRA	Government Reference Architecture
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
PVI	Pilot Vehicle Interface
ROE	Rules of Engagement
SK	Starter Kit

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not add any significant changes.	July 24, 2025	SK13.0
B	This revision did not add any significant changes.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the PVI Volume Between Documentation Releases.